

K. Nikisha

+91-7349059504 | knikisha04@gmail.com | [linkedin.com/in/nikisha-k](https://www.linkedin.com/in/nikisha-k) | github.com/nikisha77 | Bangalore, India

SUMMARY

AI and NLP Developer focused on designing intelligent automation solutions and generative AI applications. Bridging the gap between complex language models and user-facing products, with hands-on experience in vector databases and semantic search. Combines data engineering skills with a deep learning and machine learning to extract actionable insights and automate workflows.

EXPERIENCE

Business Analyst Apprentice

Trellix

July 2025 – Present

Bangalore, India

- Managed the New Product Introduction process, aligning cross-functional teams on core business requirements to elevate operational efficiency.
- Architected a Jira MCP integration and a RAG system using generative AI to automate customer escalation workflows and accelerate internal information discovery.
- Evaluated complex product metrics using Looker to construct live tracking dashboards, converting data insights into strategic product roadmap initiatives.

AI/ML Intern

CMTI

Mar 2025 – Jun. 2025

Bangalore, India

- Engineered a semantic search system for machine documentation utilizing RAG and vector embeddings, enabling natural language querying and cut document lookup time by 70%.
- Built a context-aware LLM pipeline to analyze machine downtime and OEE data, transforming raw operational log streams into instant diagnostics and cutting analysis cycles by 60%.
- Designed and deployed a real-time hardware status dashboard, providing floor operators with high-fidelity visual analytics on equipment health and factory throughput.

AI/ML Intern

Samsung R&D

Nov. 2024 – Jan. 2025

Bangalore, India

- Developed an adaptive optimization algorithm for Federated Learning that coordinates decentralized model training across edge devices, reducing training convergence time by 18%.
- Built a secure data aggregation pipeline to combine local model updates safely without exposing sensitive user training data.
- Engineered a client-filtering layer that automatically blocks corrupted or noisy gradient updates, raising final global model accuracy on the CIFAR-10 dataset by 6%.

AI Intern

Graphene Healthcare Pvt Ltd

Sep. 2024 – Oct. 2024

Bangalore, India

- Built clinical Named Entity Recognition (NER) pipeline using LLMs to isolate critical medical entities like symptoms, diagnoses, and treatment from patient raw data.
- Optimized few-shot prompting strategies and context constraints, achieving a 95% F1-score and minimizing false negatives during critical symptom parsing.
- Automated the transformation of raw clinical text into highly structured data, accelerating core data ingestion speeds for healthcare facilities.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Bash
- **AI / LLM:** LangChain, RAG, Hugging Face, Prompt Engineering, Transformers, Autonomous Agents, Vector Search (FAISS, ChromaDB)
- **ML / DL:** PyTorch, Scikit-learn, NumPy, Pandas, NLP, Generative AI, Data Analysis
- **Databases:** PostgreSQL, Neo4j (Graph DB)
- **Tools & Platforms:** Git/GitHub, Docker, Streamlit, REST APIs, Looker, Jupyter Notebook, Google Colab, Jira, Postman

PROJECTS

Automatic OEE Dashboard using LangChain

[GitHub Link](#)

Python, Streamlit, LangChain, Huggingface

- Developed an interactive Streamlit framework to analyze Overall Equipment Effectiveness (OEE) parameters across packaging pipelines with dynamic visualizations and custom granular filtering.
- Integrated an intelligent conversational interface powered by LangChain, allowing users to query live operational telemetry using natural language commands.
- Programmed a deterministic synthetic data engine to simulate realistic industrial manufacturing logs for continuous pipeline verification.

CopyCatch – Smart Plagiarism Detector using LLMs

[GitHub Link](#)

Python, LangChain, OpenAI, FAISS, Streamlit

- Formulated a multi-agent plagiarism detection system using LLMs and semantic vector indexers to cross-examine text variations, semantic rewording, and structural paraphrasing in research papers.
- Deployed a live citation cross-referencing module using Tavily and arXiv search APIs to validate background reference integrity against paper abstracts.
- Implemented AI-authored text classifiers alongside a FAISS-driven retrieval architecture, feeding comprehensive originality insights directly into a clean Streamlit portal.

EDUCATION

Ramaiah Institute of Technology

B.E. in Artificial Intelligence and Data Science | CGPA: 9.07 / 10.0

Aug. 2021 – May 2025

Bangalore, India

ACHIEVEMENTS

- **Neo4j Certified Professional:** Certified by Neo4j GraphAcademy in advanced graph data modeling, query optimization, and complex relationship mapping using Cypher.
- **Manas Codeathon (6th Place):** Designed and implemented a state-managed memory game under strict time constraints, placing 6th out of competing engineering teams to secure a Consolation Prize.
- **Ideathon @ RIT (3rd Prize):** Developed and pitched an algorithmic solution to a real-world infrastructure problem statement, securing 3rd place overall among all participating teams.